



DVM S 3 Phase Water Heat Pump
Cooling Target Operation

Water Source HP Outdoor Unit				
Component	Abbreviation		Cooling Mode	
Compressor	Comp	Operation	Target low pressure control (default: 113.8psig)	Primary function, Maintain Average Evaporator Temperature (Average temperature of Eva. In Sensors)
		Range	14~120Hz	Low pressure range 100psig - 128 psig. Evaporator temperature range 45°F -48°F
High side pressure	HP	Operation	Head pressure is effected by the water temperature	Is there a variable flow control valve installed? This will effect head pressure.
		Range	299 psig - 469 psig (water temperature 68°F-104°F)	At 228psig, low limit control is activated at 512psig high limit control is activated
Main EEV	E_M	Operation	Full Open	
		Range	2000 Steps	
EVI EEV	E_EVI	Operation	*Subcooling control (default: SC=36°F)	SC = High Side Saturation Temp. - Liquid Line Temp.
		Range	0-480step	
4 Way Valve	V_4W	Operation	Off	
EVI By-Pass Solenoid Valve	V_EB	Operation	On (Open): *T_W > 104°F or 50Hz↓	Activates Sub-Cooling Control
			Off (Close): T_W < 100.4°F↓ and 65 Hz↑	Discharge Temperature Protection, Control by unit
EVI Solenoid Valve	V_EVI	Operation	On (Open)	
Hot Gas By-Pass	V_HG	Operation	Off (Closed)	
Accumulator Oil Return Solenoid Valve	V_AR	Operation	On (Open when compressor is running)	Off during start up if DSH is < 18°F. Off for 2 min. if DSH is < 18°F On when DCSH > 27°F
Hot Gas By-Pass 2 (HR Only)	V_HG2	Operation	Off Closed	Opens after 20 minutes of continuous run time in cooling mode
Main EEV Solenoid Valve (HR Only)	V_ME	Operation	On (Open)	Energized in cooling only
Main Cooling Solenoid	VMC	Operation	Off (Closed)	
Liquid Pipe Thermistor	T_L	Operation	Used to calculate sub-cooling	Sub-Cooling range is 9°F-36°F
HP - HR Ball Valve (3 phase only)	BV_HP	Operation	Open	Valve must be open for heat pump applications
IDU				
Component	Abbreviation		Cooling Mode	Comments
Room Temperature	Room Temp.	Operation	65~86°F	
Liquid Pipe Thermistor	EVA In 1	Operation	Evaporator Saturation Temperature. Range 45-48°F	Used to calculate Evaporator Superheat
Gas Pipe Thermistor	EVA Out 1	Operation	Measures Suction Pipe Temperature	Used to calculate Evaporator Superheat
EEV	EEV 1	Operation	Modulates to maintain Superheat of 3.6 degrees F	Superheat = Pipe out Temp - Pipe in Temp. (Typical Range 0-7)
		Range	0 - 2000 Steps (0 Steps when zone is satisfied)	Typical range between 250 and 1500 Steps
IDU Fan	Fan Speed	Operation	Range	Low , Medium, High, Auto
			Low Speed: Ts - Ti ≤ 1.8°F	TS - Set Temperature
			Medium Speed: 1.8°F < Ts - Ti ≤ 5.4°F	Ti Indoor Temperature
			High Speed 5.4°F < Ts - Ti	